METHODS OVERVIEW

Procedure-agnostic post-selection inference for identified subgroups

A multiplier (wild / martingale) bootstrap recasting of the León (2024) de-biased estimator and infinitesimal-jackknife interval

forestsearch · Cox & GLM (DINA) estimands

THE PROBLEM

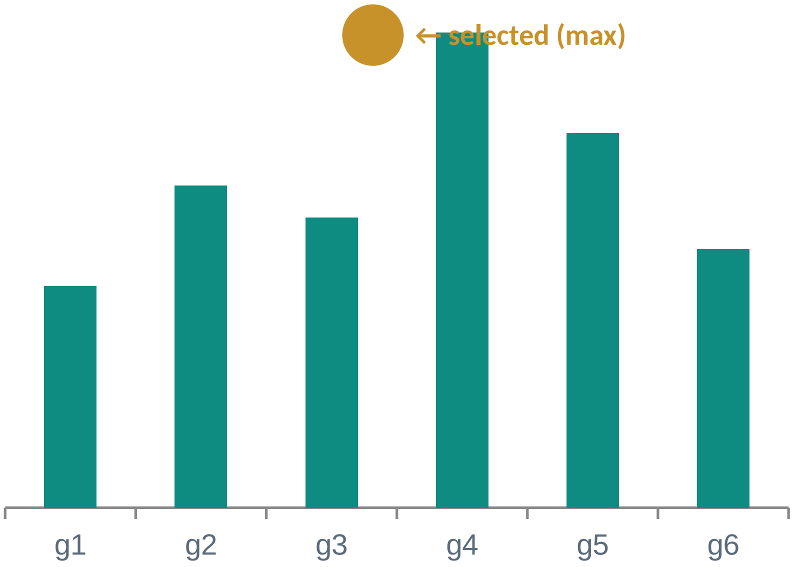
Selection optimism — the winner's curse

A subgroup chosen ***because*** its effect is extreme has an upward-biased effect. A naive within-subgroup CI, ignoring that the subgroup was selected, **under-covers.**

The inferential estimand

$\hat{\beta}(g)$ — the within-subgroup treatment effect on the natural-parameter scale (log-HR, log-OR, log-IRR, mean difference).

Candidate effects $\hat{\beta}(g)$ — the largest is selected



LEÓN ET AL. (2024)

A two-term bootstrap correction, with an IJ interval

$$\tilde{\beta}(H) = \hat{\beta}(H) - (1/B)\, \Sigma_b \left[\, \eta^*_b(H^*_b) + \eta^*_b(H) \,\right]$$

$\eta^*_b(g) = \hat{\beta}^*_b(g) - \hat{\beta}(g)$ (bootstrap minus observed effect at subgroup g)

TERM 1

Selection optimism

Discrepancy at the re-selected subgroup — the winner's curse the bootstrap re-measures on each resample.

TERM 2

Estimation bias at H

Discrepancy at the fixed observed subgroup — the estimator's own bias there, and the IJ residual.

WHAT IS NEW IN LEÓN

The bias term Harrell (1996) does not address

Harrell, Lee & Mark (1996)

Optimism correction

- **Term 1 only**
- Corrects selection optimism
- Estimator's own bias at H left uncorrected

León et al. (2024)

Two-term + IJ variance

- **Term 1 + Term 2**
- Adds $\eta^*(H)$: bias of $\hat{\beta}$ at the selected subgroup
- Makes $\tilde{\beta}$ a proper bagged estimator
- Supplies the residual for the IJ variance

THIS WORK · THE MECHANISM

A wild / martingale bootstrap of influence functions

1	**One resample = one reweighting**	Subject i gets multiplicity $K^*(b,i)$; all the randomness is one weight vector.
2	**A refit = a dot product**	$\hat{\beta}^*(g) - \hat{\beta}(g) = \Sigma\,(K^*(b,i) - 1)\cdot \mathrm{dfbeta}(g,i) = D(g)$ — no refit.
3	**One shared vector, all candidates**	Overlapping subgroups move together → faithful selection competition.
4	**Centered Poisson = the bootstrap weight**	$K^*(b,i) - 1 \to \mathrm{Poisson}(1) - 1$: the IJ linearization of the bootstrap.

→ The screen, the family routine, and the de-biasing all run from one N×S influence matrix.

THE MULTIPLIER · ONE DRAW

One weight vector holds the whole resample

$K^*(b,i)$ = the number of times subject i is drawn into resample b.

The count vector is the resample: refitting any model = reweighting the original by the counts — $\Sigma\, K^*(b,i)\cdot\psi(i) = 0$ — with no rows duplicated. One draw drives all S candidates at once.

Only the centered weight enters

1	0	2	1	1	0	2	1

counts on one draw — mostly 0 / 1 / 2 (K ≈ Poisson(1): ≈37% / 37% / 18%)

Baseline 1 = original fit; only $G = K^*(b,i) - 1$ perturbs it:
$\hat{\beta}^*(g) - \hat{\beta}(g) = \Sigma\, G\cdot\mathrm{dfbeta}(g,i) = D(g)$
G: mean 0, variance 1 → the infinitesimal jackknife.

The weight law is interchangeable

Multinomial	Poisson(1)
Rademacher ±1	Gaussian

All mean-0, unit-variance — they agree to first order. **Rademacher ±1 is the 50/50 split** (the consistency criterion); centered-Poisson = the IJ of the bootstrap.

RESULT Q1 · PROVEN

The fast estimator IS León's Eq. 7, linearized

$$\eta^*_b(g) = \Sigma_{i\in g}\,(K^*_{bi} - 1)\cdot \mathrm{dfbeta}_{g,i} = D_g(b)$$

- **Point estimate:** $\tilde{\beta}(H)$ equals León's Eq. 7 to first order — for H and its complement H^c.
- **Interval:** León's IJ variance from the same draws; wider than the naive SE (it carries selection variance).
- **Only approximation:** The dfbeta linearization — exact for mean-difference, loosest at small subgroup size.

WHAT "PROCEDURE-AGNOSTIC" MEANS

The inference wraps any selection on the effect

$$\hat{H} = \mathscr{S}(\ \{\ \hat{\beta}(g) : g \in \mathscr{F}\ \}\ )$$

TIER 1 · universal

Literal bootstrap re-run

Defined for any procedure you can re-run on a resample. The general definition of the de-biased estimator + IJ variance.

TIER 2 · fast & exact

Multiplier surrogate

First-order exact whenever the selection ranks on the inferential effect $\hat{\beta}(g)$ over an enumerable family.

THE LIMITATION

Model-based selectors must be re-ranked on the effect

GRF, DINA, and Lasso rank on their own statistic — not the Cox/GLM effect. They become candidate generators; selection moves to $\hat{\beta}(g)$.

GRF

native statistic

ranks regions by a mean doubly-robust (AIPW) score

DINA

native statistic

ranks by subgroup-mean per-patient effect $\hat{\tau} = \bar{a}\cdot\hat{\beta}_D$

Lasso

native statistic

selects a rule along a penalized coefficient path

Given up: the native ranking's own optimality as the selection criterion. **Gained:** an exactly correctable, validly covered effect.

WHY IT IS FORCED, NOT TUNING

The per-draw winner flip (worked example)

Two candidates — effect and native statistic disagree:

	g_1	g_2	selects
within-group effect $\hat{\beta}(g)$	0.90	1.00	effect → g_2
DINA subgroup-mean $\hat{\tau}$	0.55	0.50	native → g_1

Different winners on the same draw. The effect gate credits the wrong optimism for the native procedure — so its selection must rank on $\hat{\beta}(g)$.

On one resample b: D_1=+0.05, D_2=+0.30 (effect) • E_1=+0.12, E_2=+0.02 (native)

Native bootstrap (what DINA does)

T+E: g_1 0.67 > g_2 0.52 → picks g_1
credits D_1 = +0.05

Effect gate (re-selects on $\hat{\beta}$+D)

$\hat{\beta}$+D: g_1 0.95 < g_2 1.30 → picks g_2
credits D_2 = +0.30

RESULT Q2 · THE FEASIBLE TARGET

Align the criterion — exact, conditional on the family

`select_statistic = "effect"` $\Longrightarrow$ $T = \hat{\beta}$, $E = D$, one influence matrix, gate exact

What is proven

- DINA / GRF / Lasso → candidate generators
- Selection layer identical to forest search
- De-biased point + IJ interval first-order exact

The scope condition

- **Conditional on the proposed family $\mathcal{F}$**
- Exact for a fixed / enumerated family
- Data-driven $\mathcal{F}$ (DINA surface / GRF forest): family-regeneration component is the open piece

EVIDENCE IN HAND

Validated on real data; one piece left to quantify

≈637×

faster than the 500-boot bootstrap (GBSG/Cox)

1.81

de-biased HR, gate — vs 1.93 full bootstrap

0.48 / 0.36

IJ SE vs naive robust SE (log-HR)

Gate CI [0.71, 4.67] vs bootstrap [0.88, 4.24] — point estimates track; the IJ interval is mildly conservative at small n_S.

Open (Q3): under a data-driven family, the fast gate omits the family-regeneration variance. Quantify it against a literal Tier-1 that re-runs DINA/GRF per resample — the León (2024) Table 2 design, extended with a Tier-2 block and a paired agreement panel.

CONTRIBUTIONS

One correction, recast and generalized

1 **Procedure-agnostic**

León's de-biasing + IJ wrap any effect-ranked selection.

2 **Cox and GLM**

One natural-parameter (DINA) scale; one influence representation.

3 **Fast, exact to first order**

Refit-free multiplier bootstrap = León's Eq. 7 (point + IJ).

4 **A clean home for DINA/GRF**

Aligned to the effect — exact conditional on the family.

Next: simulate operating characteristics (bias · coverage · calibration) and the family-regeneration gap.